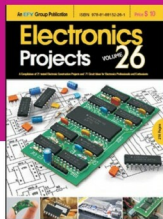


Electronics Projects Vol. 26 is the latest volume in the series, available in digital form.

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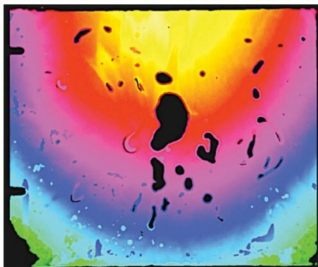


Fig. 4: Colours indicating specific depths map the warpage in the ceramic raft of an IGBT module. Black items (and smaller coloured items) are voids in the solder

and can create images at any depth, including the die attach near the top of the module.

Besides internal gaps, there is another reason to image IGBT modules acoustically: The heat-sink is bonded to a ceramic raft, which tends to warp or tilt. Even if there are no voids in the adhesive, a distorted raft makes heat flow from the die non-uniformly, and creates mechanical stresses within the die that can lead to fracture and electrical failure of the die.

Both anomalies—voids and raft distortion—are imaged by the acoustic microscope tool in a single scan of the module by using the Time Difference mode. The return echoes in this case do not report the amplitude of the echo. Instead, these report the distance of the reflecting surface from a reference point. The gate defining the depth range from which echoes will be accepted extends from just below the interface between the solder and the heat-sink to a depth just below the deepest point on the surface of the ceramic raft. The gate itself is planar, so the thousands or millions of echoes collected from x-y coordinates will create a surface contour map of the raft.

Fig. 4 shows the Time Difference image of one of nine ceramic rafts in an IGBT module. The colour spectrum maps the surface of the warped ceramic raft: the deepest points (nearest to the die) are yellow. Red, pink and subsequent colours are closer to the heat-sink. The black regions in the lower left and right corners are

either in contact with the heat-sink or very close to it—that is, these regions are above the gate and imaged as black acoustic shadows.

Numerous voids of various sizes are also present in the solder. The black ones are above the gate and adjacent to the heat-sink; others are at the depth defined by their colour.

The die attach material above the raft in an IGBT module is a second location where voids may occur, and the acoustic microscope tool scanning the heat-sink can also image the die attach. The return echoes from the die attach must pass through the solder layer, with the result that voids in the solder (like those in Fig. 4) may hide some of the voids in the die attach. But an IGBT module with voids at both locations is likely to be rejected anyhow, although it can be reworked if not yet encapsulated.

Acoustic microscope tools have numerous other imaging modes that are useful in examining heat-sinks—non-destructive cross-sectioning, for example, or the ability to simultaneously make dozens of very thinly gated images at specific depths. In all cases, the goal is to image and analyse features that might interfere with heat flow. **EFY**